OVUM G-1

Engineered from the Broken



Metodology

Topology optimization at critical points to generate shapes in Fusion using generative design tools.

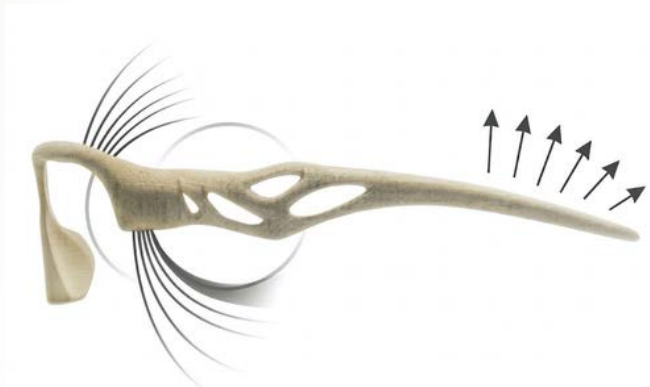
Objective

To transform organic shell waste into high-performance bioceramics material, developing ultra-lightweight, sustainable, and fully circular optical frames.

Ergonomy

This frame combines generative design and anatomical principles to minimize weight and prevent fatigue. Its wraparound geometry enhances peripheral vision, while the optimized fit at the nose bridge maximizes comfort. Additionally, it incorporates ventilation channels and high-grip design to ensure the eyewear remains securely in place and vision stays unobstructed.



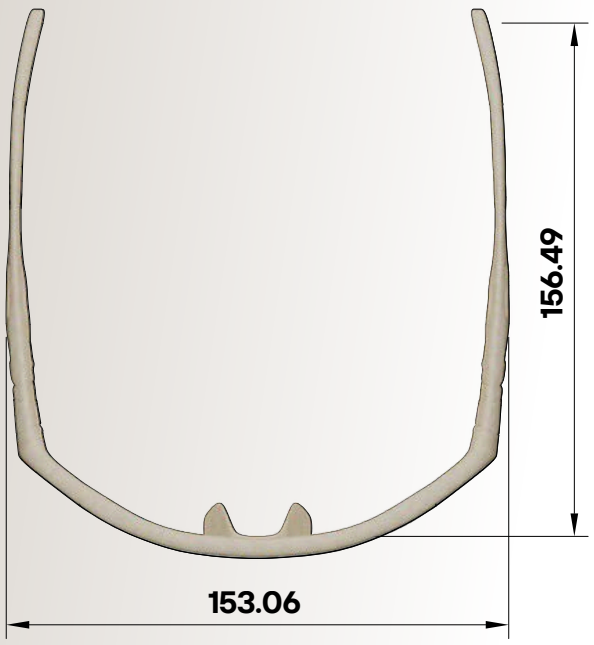


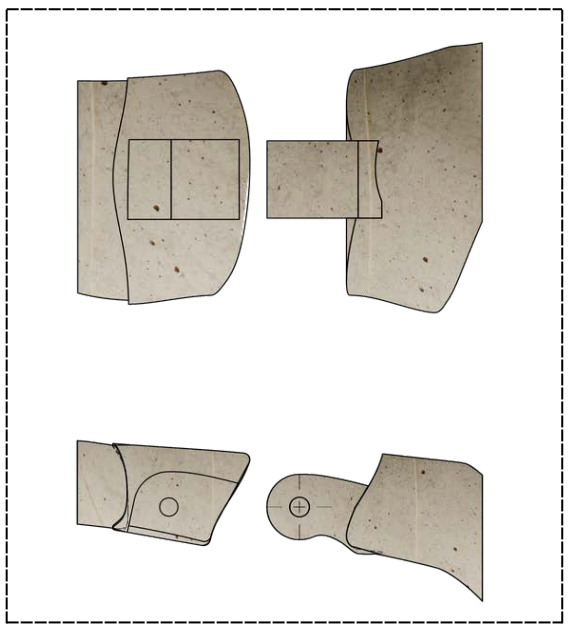




Lens

The system utilizes a mechanical snap-fit connection from the base of the frame, eliminating the need for screws. This structural geometry ensures secure retention and enables quick and easy assembly or lens replacement.





Interlocking components

Hinges

The hinges are engineered to absorb cyclic loading and protect the structural integrity of the organic material. The mechanism distributes stress evenly across the temples, ensuring smooth articulation and extending the lifespan of the assembly through a stable fit.





Material

The eggshell paste transforms organic waste into a structurally sound material, exemplifying circular economy principles. Its natural, textured finish eliminates artificial elements, directly linking the final product to its biological roots.



Form

The frame applies functional minimalism, where the geometry is streamlined to reduce visual impact and bulk. The design seeks maximum structural integrity and ergonomics through minimal material usage, eliminating superfluous elements to prioritize lightweight performance and the pure essence of the object.



Biomimetics

The temple design leverages topology optimization, emulating natural structures such as trabecular bone or mycelium networks. Rather than a mere aesthetic feature, this is a strategic subtraction of material in low-stress regions. The result is a highly rigid and mechanically robust component with significantly reduced mass, achieving a true convergence of biomimicry and engineering efficiency